

CS3423 Mini Assignment 2

Q2: Shift-Reduce and Reduce-Reduce Conflicts

CS24BTECH11030, CS24BTECH11036

1 Shift-reduce conflict — `sr_conflict.y`

```
%token IF ELSE EXPR STMT

%%
stmt:
    IF expr stmt
  | IF expr stmt ELSE stmt
  | STMT
  ;

expr:
    EXPR
  ;

%%
```

This is the classic *dangling-else* grammar. After the parser has seen `IF expr stmt` and the lookahead token is `ELSE`, it can either:

- **shift** the `ELSE` (starting to match the second alternative, attaching the `else` to the closest unmatched `if`), or
- **reduce** `IF expr stmt` to `stmt` using the first alternative (attaching the `else` to an outer `if`).

The grammar itself does not say which one to do, so the parser generator has to choose one of the two actions for the same state/lookahead pair – this is a **shift-reduce conflict**. Bison resolves it by shifting (which is also what real languages want – an `else` binds to the nearest `if`), but it still reports the ambiguity.

Compiling with bison gives:

```
$ bison -y -v sr_conflict.y
sr_conflict.y: warning: 1 shift/reduce conflict [-Wconflicts-sr]
sr_conflict.y: note: rerun with option '-Wcounterexamples' to
                    generate conflict counterexamples
```

2 Reduce-reduce conflict — `rr_conflict.y`

```
%token ID
```

```

%%
stat:
    A
  | B
  ;

A:
    ID
  ;

B:
    ID
  ;
%%

```

Here both nonterminals A and B are defined as exactly the single token ID, with no other context to tell them apart. After the parser sees an ID, it must decide whether to reduce it to A or to B – both reductions are valid at that point, so this is a **reduce-reduce conflict**. Bison breaks the tie by preferring the rule that appears earlier in the grammar (A), which makes the B: ID rule permanently unreachable.

Compiling with bison gives:

```

$ bison -y -v rr_conflict.y
rr_conflict.y: warning: 1 reduce/reduce conflict [-Wconflicts-rr]
rr_conflict.y: note: rerun with option '-Wcounterexamples' to
                    generate conflict counterexamples
rr_conflict.y:14.7-8: warning: rule useless in parser due to
                    conflicts [-Wother]

14 | ID
   | ~~

```

3 Files submitted

- `sr_conflict.y` — dangling-else grammar, 1 shift/reduce conflict.
- `rr_conflict.y` — two identical single-token rules, 1 reduce/reduce conflict.

Both were verified with `bison` (version 3.8.2, run as `bison -y -v`) and reproduce the exact warnings shown above.